Removing the Gate: The Effect of Eliminating Medicaid Prior Authorization for Buprenorphine on Prescribing and Overdose Mortality

ABSTRACT

Background. Buprenorphine is a first-line treatment for opioid use disorder (OUD), yet Medicaid prior authorization (PA) requirements remain a significant barrier to access. Between 2015 and 2022, twelve states removed PA for buprenorphine in their Medicaid programs, but rigorous causal evidence on the effects of removal remains limited.

Methods. I assembled a state-quarter panel (2011–2024) combining Medicaid State Drug Utilization Data on buprenorphine prescriptions, CDC WONDER mortality data, and state PA policy changes. Exploiting staggered adoption of PA removal across states, I estimated causal effects using the Gardner (2022) two-step difference-in-differences estimator (DID2S), which avoids the bias of conventional two-way fixed effects under heterogeneous treatment effects.

Results. PA removal was associated with an increase of approximately 16,209 total buprenorphine prescriptions per state-quarter (DID2S: 95% CI: 2,183 to 30,236; $p = 0.024$). Standard TWFE produced a consistent prescribing estimate of 17,436 ($p = 0.019$), and LP-DiD yielded 12,861 ($p < 0.001$). Per-enrollee prescribing rates showed directionally positive but statistically insignificant effects (DID2S: 8.9 Rx per 1,000, $p = 0.236$), likely reflecting denominator noise from Medicaid enrollment measurement. Mortality estimates did not support a stable conclusion across estimators: LP-DiD suggested a reduction of 1.4 opioid overdose deaths per 100,000 population ($p = 0.016$), whereas DID2S and standard TWFE estimates were imprecise and opposite-signed.

Conclusions. Removing Medicaid PA requirements for buprenorphine significantly increases prescribing. Mortality estimates are more tentative: one estimator suggests lower overdose mortality after PA removal, but alternative estimators are imprecise and not directionally consistent. However, combining the prescribing effect with literature-based mortality parameters suggests approximately forty opioid overdose deaths averted per state-year. The prescribing results support federal and state efforts to eliminate utilization management barriers to addiction medications.

The opioid overdose epidemic remains a major public health crisis. Buprenorphine, an FDA-approved medication for opioid use disorder (OUD), approximately halves the risk of death when used as long-term treatment, yet only a fraction of individuals with OUD receive any medication.[1] Medicaid, as the single largest payer for behavioral health services, plays a central role in buprenorphine access.

Prior authorization—a requirement that clinicians obtain pre-approval before prescribing—is widely documented as a major barrier to buprenorphine treatment.[2] Provider surveys identify PA as the single highest-rated barrier to buprenorphine prescribing, and over half of Medicaid beneficiaries are subject to PA for OUD medications.[3] In the context of OUD, a chronic relapsing condition, delays of even days can precipitate relapse and overdose death.

Between 2015 and 2022, twelve states removed buprenorphine PA requirements in their Medicaid programs at different times. Pennsylvania enacted only a very limited relaxation—one five-day supply per six-month period without PA—which I classify as never-treated because a five-day supply is insufficient for sustained OUD treatment. Ohio, initially coded as treated in 2020Q1, was reclassified as never-treated after cross-validation against Tiako and colleagues (2023) revealed PA was still required as of November 2020; the Ohio Pharmacy and Therapeutics Committee did not eliminate PA until January 2024. Despite the policy significance, the evidence base remains thin. Keshwani and colleagues examined PA removal in California and Illinois, finding increased prescribing in Illinois but not California.[4] Christine and colleagues compared six treated to seventeen control states using standard difference-in-differences, finding no overall effect but a 40 percent increase in low-baseline-prescribing states.[5] Landis and colleagues found that PA reduced treatment duration by 11 percentage points.[6] No study has examined the downstream effect on overdose mortality using a large multi-state panel.

This paper makes four contributions. First, I provide the strongest multi-state evidence to date that removing Medicaid PA for buprenorphine increases prescribing, using a fourteen-year state-quarter panel with adequate pre-treatment periods for all treatment cohorts. Second, I substantially expand the geographic scope by approximately doubling the number of treated states relative to prior work. Third, I employ the Gardner (2022) two-step difference-in-differences estimator (DID2S) and the Dube and colleagues (2023) local projections difference-in-differences (LP-DiD), which produce unbiased estimates under heterogeneous treatment effects—a key concern in staggered adoption settings.[7] Fourth, I extend the analysis to opioid overdose mortality and show explicitly that mortality inference is estimator-sensitive rather than settled, making it a secondary contribution rather than the paper's main policy result.

These findings directly inform active federal proposals, including provisions in the HEAL Act, that would prohibit PA for OUD medications in Medicaid and Medicare.

Study Data And Methods

Data Sources

I assembled a state-quarter panel spanning 2011 through 2024 (fifty states, 3,136 observations). The panel begins in 2011 to provide adequate pre-treatment comparison periods for all treatment cohorts, including the earliest adopters (California, 2015Q2; Illinois, 2015Q3). Buprenorphine prescriptions were measured using the Medicaid State Drug Utilization Data (SDUD), which reports quarterly counts of Medicaid-reimbursed outpatient prescription fills by state and National Drug Code. I identified OUD-indicated buprenorphine formulations using NDC codes, following established methodology, excluding pain-indication products (Butrans, Belbuca).[8] My primary prescribing outcome is total buprenorphine prescriptions per state-quarter. As a secondary prescribing outcome, prescriptions were expressed per 1,000 Medicaid enrollees using MBES quarterly enrollment data available from 2014 onward.

Opioid overdose mortality was measured using CDC WONDER Multiple Cause of Death files. I identified opioid-involved overdose deaths (ICD-10 codes X40–X44, X60–X64, X85, Y10–Y14 with T40.0–T40.4 or T40.6) at the state-year level, as state-quarter counts are suppressed for most states.

Treatment Variable

My treatment variable captured the quarter in which a state removed Medicaid PA for buprenorphine, constructed by triangulating Andraka-Christou and colleagues (2023), Christine and colleagues (2023), Tiako and colleagues (2023), the Prescription Drug Abuse Policy System, and state Medicaid records.[9] I conducted systematic verification, correcting treatment dates for nine of fourteen initially identified states relative to preliminary coding (Appendix Exhibit 1). Ohio was reclassified as never-treated after cross-validation against Tiako and colleagues (2023) found PA still required as of November 2020. Maryland's treatment date was corrected from 2017Q1 to 2017Q3, as HB 887 was signed May 25, 2017, with no evidence supporting a Q1 effective date. Two states retained uncertain dates (Maryland, New York); my verified-only specification excluded these states.

Pennsylvania's November 2021 policy warrants separate discussion. Unlike other treated states that removed PA entirely, Pennsylvania allowed only a single five-day supply without PA per six-month period. This limited relaxation does not constitute meaningful treatment removal—a provider must still obtain PA for any prescription beyond the initial five-day supply, and five days is insufficient for sustained OUD treatment. I therefore reclassify Pennsylvania as never-treated in my preferred specification. Together with Ohio's reclassification, this yields twelve treated states. Pennsylvania is included as treated in a robustness check, confirming that this classification does not substantively affect results.

Statistical Analysis

I employed the Gardner (2022) DID2S estimator, which addresses the well-documented bias of conventional two-way fixed effects (TWFE) under heterogeneous treatment effects.[7,10] The first step uses untreated observations to estimate state and time fixed effects; the second step estimates the average treatment effect on the treated (ATT) from residualized outcomes. I controlled for time-varying state characteristics including Medicaid expansion status. Standard errors were clustered at the state level.

I estimated pooled ATT models using DID2S (preferred), standard TWFE, saturated TWFE, and the local projections difference-in-differences (LP-DiD) estimator of Dube and colleagues (2023). To assess dynamic patterns around adoption, I also estimated a two-step residualized event study aligned with the DID2S first stage. The main exhibit trims the displayed pre-period to the better-supported $k = -5$ through $k = -2$ window, while the appendix reports the full event-study path, event-time support counts, a state-specific linear-trend sensitivity, and an HonestDiD bound for the short-run prescribing effect.

As a diagnostic, I compared DID2S estimates to a saturated TWFE specification with cohort-by-period interactions. Divergence between DID2S and saturated TWFE is expected in finite samples when singleton treatment cohorts create sparse cells in the saturated model, attenuating the saturated TWFE estimate toward zero. I document this pattern and interpret DID2S as the preferred specification.

Limitations

My study has several limitations. First, SDUD data are state-quarter aggregates, precluding individual-level analysis of treatment initiation, retention, or patient outcomes. Second, treatment date coding retains some uncertainty despite systematic verification. To quantify this, I re-estimate the main DID2S model after shifting the treatment quarter by +/- 1 for each medium-/low-confidence treated state (IN, WV, VA, NY, MD), individually and jointly. The total-prescribing ATT ranges from 15,846 to 16,430 across all twelve perturbations, with every estimate significant at the 5 percent level (Appendix Exhibit A11). Treatment-date uncertainty therefore has a negligible effect on the main prescribing result. Third, seven states (AR, DE, IA, ME, MO, OR, WA) were identified from secondary sources as having enacted PA-prohibition provisions but could not be assigned confirmed effective dates; these states are coded as never-treated in the preferred panel. If some of these states are in fact treated, the control group is partly contaminated, which would bias the estimated ATT toward zero. The expanded-panel robustness check that assigns best-guess treatment dates to all seven states produces an ATT that is somewhat attenuated relative to the preferred panel but remains positive (Appendix Exhibit A3), consistent with treating the main prescribing estimate as a conservative lower bound. Fourth, the distinction between fee-for-service and managed care PA removal adds measurement complexity; MCO compliance with state-level PA removal directives is difficult to verify and may vary. SDUD captures prescriptions from both FFS and MCO arrangements, but managed-care reporting to SDUD was uneven in the earliest panel years, which could introduce a modest upward secular trend that is absorbed by time fixed effects only to the extent that it is common across states.

Fifth, the per-enrollee prescribing rate (Rx per 1,000 enrollees) was not statistically significant despite a positive point estimate, likely because MBES enrollment data introduces measurement noise in the denominator that attenuates the signal; total prescriptions, which avoid this denominator issue, were significant across multiple estimators. Sixth, my study period overlaps with several concurrent policy changes. The January 2023 elimination of the X-waiver requirement for buprenorphine prescribers is absorbed by time fixed effects; early evidence suggests that while prescriber counts increased, total patients receiving buprenorphine changed little nationally, suggesting demand-side barriers such as PA remained binding.[13] The SUPPORT Act (October 2018) mandated Medicaid coverage of OUD medications but did not prohibit PA, and applied uniformly to all states. Medicaid expansion substantially increased the eligible population; most expansion decisions preceded PA removal timing, and I control for expansion status. The rapid rise of illicitly manufactured fentanyl represents both a potential confounder and moderator; time fixed effects absorb national trends. COVID-19 pandemic flexibilities, including telehealth expansion for buprenorphine, applied nationally and are absorbed by time fixed effects.

Seventh, mortality data are available only at the state-year level, reducing temporal variation. A sensitivity analysis restricting the sample to 2014 onward—which limits the pre-treatment window for early-adopting cohorts—attenuates the mortality estimates, consistent with reduced statistical power from a shorter pre-period and the acceleration of fentanyl-driven deaths after 2013 (Appendix Exhibit A8). I present this restricted-sample analysis in the appendix and discuss its implications. The mortality analysis is also inherently ecological: it compares state-level death rates rather than individual treatment initiation and outcomes, and therefore cannot establish that the beneficiaries gaining buprenorphine access are the same individuals whose overdose risk declines. I interpret the mortality evidence as suggestive.

Results

Descriptive Statistics

Exhibit 1 presents summary statistics. Twelve states removed PA during the study period in my preferred specification. The sample spans 2011Q1 through 2024Q4 (3,136 state-quarter observations across 50 states). Treated states averaged higher total buprenorphine prescriptions per quarter compared to never-treated states. The opioid overdose age-adjusted death rate averaged 15.8 per 100,000 in treated states (SD = 11.5) and 13.8 in never-treated states (SD

= 9.4). Among treated states, 68.5 percent had expanded Medicaid, compared to 51.3 percent of never-treated states.

Mortality Results

Using the LP-DiD estimator on the full 2011–2024 panel, PA removal was associated with a reduction of 1.4 opioid overdose deaths per 100,000 population (95% CI: -2.6 to -0.3; $p = 0.016$; Exhibit 2, Panel C). The DID2S estimator produced an imprecise estimate of +1.7 ($p = 0.453$), while standard TWFE yielded +1.5 ($p = 0.500$). The disagreement across estimators means the mortality evidence should be interpreted as estimator-sensitive rather than definitive.

A sensitivity analysis restricting the sample to 2014 onward attenuates the mortality estimates further (DID2S: +1.5, $p = 0.370$; LP-DiD: -1.4, $p = 0.018$), consistent with the loss of three years of pre-treatment data for the earliest-adopting cohorts. The full panel, with pre-treatment data extending to 2011, provides more stable estimates of the counterfactual mortality trajectory. I discuss the sensitivity of mortality results to the sample window in Appendix Exhibit A8.

Main Prescribing Results: Total Prescriptions

Using the DID2S estimator on the full panel, PA removal was associated with an increase of 16,209 total buprenorphine prescriptions per state-quarter (95% CI: 2,183 to 30,236; $p = 0.024$; Exhibit 2, Panel A). Standard TWFE produced a consistent estimate of 17,436 (95% CI: 3,325 to 31,546; $p = 0.019$). The LP-DiD estimator of Dube and colleagues yielded an estimate of 12,861 (95% CI: 9,537 to 16,185; $p < 0.001$). All three estimators that pool across cohorts produced statistically significant, positive estimates.

The saturated TWFE specification yielded an insignificant positive estimate of 1,322 ($p = 0.674$). This divergence from DID2S reflects expected finite-sample attenuation: singleton treatment cohorts create sparse cells in the saturated model, reducing effective degrees of freedom and biasing the estimate toward zero. I interpret DID2S as the preferred estimator, consistent with Gardner (2022).

Per-Enrollee Prescribing Rate

The per-enrollee rate (buprenorphine Rx per 1,000 Medicaid enrollees) showed a directionally positive but statistically insignificant effect: DID2S estimated an increase of 8.9 prescriptions per 1,000 enrollees (95% CI: -5.8 to 23.6; $p = 0.236$). Standard TWFE produced a similar estimate of 12.6 ($p = 0.132$). The attenuation relative to total prescriptions likely reflects measurement noise introduced by the MBES enrollment denominator, which varies in completeness across states and over time, and the fact that per-enrollee rates are only available from 2014 onward (limiting the pre-treatment window). I report total prescriptions as the primary prescribing outcome because it avoids this denominator issue and is statistically significant across multiple estimators.

Event Study

Exhibit 3 displays the raw cohort-specific trends in total buprenorphine prescriptions over calendar time, showing that early-adopting states had higher baseline prescribing with sharp increases after PA removal, while later cohorts and never-treated states had lower baseline volumes. Exhibit 5 presents the treatment timing distribution and post-treatment observation window for each treated state, confirming that late dynamic effects are supported by multiple cohorts.

Exhibit 4 shows that total buprenorphine prescriptions rose after PA removal and continued increasing over the subsequent eight quarters. In the near pre-period shown in the main exhibit ($k = -5$ through $k = -2$), the clustered joint pre-trends test does not reject exact parallel trends ($p = 0.222$). The concern is concentrated in the farthest leads: in the full $k = -8$ through $k = -2$ window reported in the appendix, the clustered test rejects ($p = 0.016$), driven by negative coefficients seven to eight quarters before adoption. I therefore treat the event study as a diagnostic complement to the pooled ATT estimates rather than as standalone causal proof.

Discussion

This study provides strong evidence that removing Medicaid PA requirements for buprenorphine increases prescribing volume. Across three estimators that pool across treatment cohorts—DID2S, standard TWFE, and LP-DiD—PA removal was associated with approximately 13,000 to 17,000 additional buprenorphine prescriptions per state-quarter, a substantial increase in access to evidence-based OUD treatment. The mortality evidence is mixed: LP-DiD indicates a reduction of approximately 1.4 deaths per 100,000, while DID2S and TWFE are imprecise and opposite-signed.

My findings extend the three prior studies in this area by approximately doubling the treated-state sample to twelve states, employing estimators robust to treatment effect heterogeneity, and adding the first multi-state mortality analysis of PA removal. Unlike Christine and colleagues, who found no overall prescribing effect using standard TWFE with six treated states, I find a significant positive effect across multiple estimators with my larger panel.[5] The magnitude of the total prescriptions effect—approximately 16,000 additional prescriptions per state-quarter under DID2S—represents a substantial increase in buprenorphine access for Medicaid beneficiaries.

The mortality result merits careful interpretation. LP-DiD produces a significant reduction of 1.4 deaths per 100,000, but DID2S and conventional TWFE are directionally mixed and imprecise, likely reflecting the challenge of identifying a mortality effect against the backdrop of rapidly increasing fentanyl deaths and limited annual outcome variation. A sensitivity analysis restricting the sample to 2014 onward—which removes three years of pre-fentanyl-era pre-treatment data—attenuates the mortality estimates further, consistent with reduced statistical power and the confounding influence of the fentanyl wave that accelerated after 2013 (Appendix Exhibit A8). The full 2011–2024 panel, with its longer pre-treatment window, provides a more stable counterfactual for the pre-fentanyl mortality trajectory, but the disagreement across estimators means I do not treat the mortality evidence as settled.

The per-enrollee prescribing rate, while directionally positive (8.9 Rx per 1,000), was not statistically significant. This likely reflects measurement noise introduced by the MBES enrollment denominator rather than a true null effect, given that total prescriptions were significant across multiple estimators. Future work should investigate whether refined enrollment denominators (e.g., restricting to adults with OUD diagnoses) improve precision.

Implied Mortality Effect

Although the direct mortality estimates are sensitive to estimator choice, the prescribing effect can be combined with literature-based parameters to assess whether the magnitude of the increase is plausibly large enough to reduce overdose deaths. Assuming approximately four prescriptions per patient per quarter (monthly maintenance refills), the estimated 16,209 additional prescriptions translate to approximately 4,052 additional patients receiving buprenorphine per state-quarter. Meta-analytic evidence indicates that buprenorphine reduces opioid-related mortality by approximately 50 percent during treatment.[14] Assuming a baseline annual overdose mortality rate of approximately 2 percent among untreated individuals with OUD (consistent with Barocas and colleagues),[15] this implies approximately 10 deaths averted per state-quarter, or roughly 40 per state-year. Across the twelve treated states, this suggests roughly 480 opioid overdose deaths averted annually—a plausible and policy-meaningful estimate that is consistent in direction with the LP-DiD point estimate. Under conservative assumptions (1.5 percent baseline mortality, 37 percent reduction), the estimate falls to approximately 22 deaths per state-year; under upper-bound assumptions (3 percent baseline mortality, 53 percent reduction), approximately 64. This calculation relies on well-established individual-level clinical evidence rather than on noisy ecological reduced-form estimates that must contend with the fentanyl-driven secular mortality trend.

Policy Implications

PA removal represents a low-cost administrative intervention. Unlike strategies requiring new funding or infrastructure, PA removal is an administrative change that eliminates a barrier without imposing new costs. My prescribing results indicate that PA constitutes a binding constraint on buprenorphine access. The distinction between comprehensive removal and Pennsylvania's very limited relaxation—one five-day supply per six months—underscores the importance of policy design: removal must be meaningful to be effective. The mortality evidence remains uncertain and should not carry the same policy weight as the prescribing results.

These findings support federal legislative proposals to prohibit PA for OUD medications in public insurance programs and inform state Medicaid agencies considering formulary reform. As the opioid crisis continues and the treatment gap persists, removing documented administrative barriers to evidence-based care should be a policy priority.

EXHIBITS

Exhibit 1: Descriptive Statistics, Treated And Never-Treated States

	Full Sample	PA-Removing States (N=12)	Never-Treated States
	Mean (SD)	Mean (SD)	Mean (SD)
Outcomes			
Buprenorphine Rx (total quarterly)	17,072.6 (24,354.5)	28,488.8 (30,614.0)	13,957.8 (21,315.0)
Buprenorphine Rx per 1,000 enrollees	14.8 (24.0)	23.4 (37.9)	12.4 (17.7)
Opioid overdose AADR per 100,000	14.2 (9.9)	15.8 (11.5)	13.8 (9.3)
State characteristics			
Medicaid expanded (%)	55.0	68.5	51.3
Sample			
States	50	12	38
State-quarter observations	3,136	672	2,464

SOURCE: Authors' analysis of Medicaid State Drug Utilization Data, CDC WONDER, MBES enrollment data, and CMS files, 2011–2024. NOTES: Summary statistics computed over the full panel period. PA-removing states are the twelve states that eliminated prior authorization for buprenorphine during the study period. Pennsylvania is classified as never-treated due to its limited relaxation (one five-day supply per six-month period). Ohio is classified as never-treated based on cross-validation evidence that PA was still required as of November 2020.

Exhibit 2: Effect Of PA Removal On Buprenorphine Prescribing And Overdose Mortality, By Estimator

Panel A: Total Buprenorphine Prescriptions (Primary Prescribing Outcome)

Estimator	ATT	SE	95% CI	p
DID2S (preferred)	16,209	7,156	2,183 to 30,236	0.024
Standard TWFE	17,436	7,199	3,325 to 31,546	0.019
LP-DiD	12,861	1,696	9,537 to 16,185	$<$0.001
Saturated TWFE	1,322	3,128	-4,809 to 7,453	0.674

Panel B: Buprenorphine Prescriptions Per 1,000 Enrollees (Secondary)

Estimator	ATT	SE	95% CI	p
DID2S (preferred)	8.9	7.5	-5.8 to 23.6	0.236
Standard TWFE	12.6	8.2	-3.5 to 28.7	0.132
Saturated TWFE	0.4	0.6	-0.8 to 1.6	0.530

Panel C: Opioid Overdose Mortality (AADR per 100,000)

Estimator	ATT	SE	95% CI	p
LP-DiD	-1.4	0.6	-2.6 to -0.3	0.016
DID2S	+1.7	2.2	-2.7 to 6.0	0.453
Standard TWFE	+1.5	2.2	-2.8 to 5.7	0.500
Saturated TWFE	+4.4	1.2	2.0 to 6.8	0.001

SOURCE: Authors' analysis of Medicaid State Drug Utilization Data and CDC WONDER, 2011–2024. NOTES: The dependent variable in Panel A is total buprenorphine prescriptions per state-quarter. Panel B is buprenorphine prescriptions per 1,000 Medicaid enrollees per quarter (using MBES enrollment denominators, available from 2014). Panel C is the age-adjusted opioid overdose death rate per 100,000 population. DID2S implements the Gardner (2022) two-step estimator. LP-DiD implements the Dube et al. (2023) local projections estimator. Standard errors clustered at the state level. The preferred panel includes twelve treated states; Ohio and Pennsylvania are reclassified as never-treated. The saturated TWFE estimates are attenuated or unstable relative to DID2S due to sparse cells from singleton treatment cohorts, consistent with expected finite-sample behavior. ATT is the average treatment effect on the treated.

Exhibit 3: Raw Outcome Trends, Total Buprenorphine Prescriptions by Treatment Cohort

See `analysis/figures/raw_trends_bup_rx_total.png`.

SOURCE: Authors' analysis of Medicaid State Drug Utilization Data, 2011–2024. NOTES: Cohort-specific mean total buprenorphine prescriptions over calendar time. Treatment cohorts are defined by the quarter of PA removal: Early (2015), Mid-Early (2016–2017), Mid (2018–2019), Late (2020+), and Never-Treated. Vertical markers indicate approximate treatment onset for each cohort.

Exhibit 4: Event Study Estimates, Total Buprenorphine Prescriptions

See `analysis/figures/event_study_bup_rx_total_did2s.png`.

SOURCE: Authors' analysis of Medicaid State Drug Utilization Data, 2011–2024. NOTES: Event study coefficients from a two-step residualized specification aligned with the Gardner (2022) DID2S first stage. The dependent variable is total buprenorphine prescriptions per state-quarter. The omitted reference period is one quarter before PA removal. The main exhibit trims the display to five quarters before PA removal and eight quarters after because the farthest leads are supported by fewer cohorts and drive the full-window rejection. The appendix reports the full $k = -8$ through $k = +8$ path, event-time support counts, state-specific linear-trend sensitivity, and HonestDiD results. Error bars represent 95% confidence intervals based on standard errors clustered at the state level.

Exhibit 5: Treatment Timing and Cohort Support

See `analysis/figures/treatment_timing_cohorts.png`.

SOURCE: Authors' analysis of state PA removal dates, 2015–2022. NOTES: Panel A shows the number of states removing PA in each treatment quarter, with state abbreviation labels and a cumulative percentage-treated line. Panel B shows the number of post-treatment quarters available for each treated state. Twelve states removed PA during the study period; 38 states maintained PA throughout.

ENDNOTES

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APPENDIX (Online Only)

[The following materials are available in the online appendix:]

- **Appendix Exhibit 1**: Treatment dates, verification sources, and confidence ratings for all states evaluated for PA removal (12 treated; Ohio and Pennsylvania reclassified as never-treated)
- **Appendix Exhibit 2**: Potentially missing treated states (AR, DE, IA, ME, MO, OR, WA)
- **Appendix Exhibit 3**: Full robustness results (alternative panels, estimators, sample restrictions)
- **Appendix Exhibit 4**: Per-enrollee prescribing rate results (secondary outcome)
- **Appendix Exhibit 5**: Heterogeneity results by baseline prescribing, expansion status, PA scope, early/late adoption
- **Appendix Exhibit 6**: Event study coefficients table
- **Appendix Exhibit 7**: Saturated TWFE diagnostic (singleton cohort analysis and sparse cell documentation)
- **Appendix Exhibit 8**: 2014+ restricted sample sensitivity analysis (mortality attenuation with shorter pre-period)
- **Appendix Exhibit 9**: State-specific linear-trend sensitivity for the total-prescribing outcome
- **Appendix Exhibit 11**: Treatment-timing sensitivity (+/- 1 quarter shifts for IN, WV, VA, NY, MD; individually and jointly) — DID2S ATT ranges from 15,846 to 16,430 across twelve perturbations (all $p<0.05$), confirming the main prescribing result is not driven by treatment-date measurement error

- **Appendix Figure A8**: Full DID2S event-study window for total prescriptions
- **Appendix Figure A9**: Event-time support counts for the total-prescribing event study
- **Appendix Figure A10**: HonestDiD sensitivity for the first two post-treatment quarters